08-14 第1篇

阅读理解（说明文）

① The Mariana Trench (马里亚纳海沟) is the deepest place in the world's oceans. It lies in the western Pacific Ocean, east of the Philippines and south of Japan. The trench is about 2,550 kilometers long and 69 kilometers wide at its broadest point. Its deepest point, known as the Challenger Deep（挑战者深渊）, reaches about 11 kilometers below sea level. If you placed Mount Everest (珠穆朗玛峰) at the bottom, its peak would still be over 2 kilometers underwater.

② Why does the Mariana Trench exist? It was formed by a process called subduction (俯冲). The Pacific Plate (太平洋板块) slowly moves under the Philippine Sea Plate. When one tectonic plate (构造板块) slides under another, it creates a deep valley in the ocean floor. This process has been going on for millions of years.

③ The conditions in the trench are extreme. There is no sunlight at such depths. The water pressure is enormous — more than 1,000 times greater than at sea level. The water is very cold, only about 1 to 4 degrees Celsius. Yet, even in this harsh environment, some life forms survive. Scientists have found fish, shrimp (虾), and other strange deep-sea creatures that have adapted to the darkness and pressure.

④ Exploring the Mariana Trench is very difficult and expensive. In 1960, two men — Jacques Piccard and Don Walsh — became the first people to reach the Challenger Deep in a special submarine. They stayed at the bottom for about 20 minutes. In 2012, film director James Cameron made a solo dive to the same place. Today, robots and remote-controlled vehicles (遥控潜水器) help scientists study the trench without sending humans down.

⑤ The Mariana Trench holds many secrets. Scientists believe studying it can help us understand how life began on Earth and how our planet works. Protecting this unique place is also important as human activities may one day affect even the deepest parts of the ocean.

注释： trench 海沟；subduction 俯冲；tectonic 构造的；plate 板块；enormous 巨大的；harsh 严酷的；submarine 潜水艇；solo 独自的

选择题

1. Where is the Mariana Trench located? A. In the Atlantic Ocean. B. East of the Philippines and south of Japan in the Pacific. C. Near the coast of China. D. In the Indian Ocean.

2. How deep is the Challenger Deep? A. About 2,550 kilometers. B. About 11 kilometers below sea level. C. About 69 kilometers. D. About 2 kilometers below sea level.

3. According to paragraph ③, what is true about life in the Mariana Trench? A. There are no living things there at all. B. Sunlight reaches the bottom of the trench. C. Some creatures have adapted to the darkness and pressure. D. The water temperature is very high.

4. What can we infer from paragraph ④ about exploring the trench? A. It is easy and cheap for anyone to visit. B. Only one person has ever reached the Challenger Deep. C. Scientists no longer use submarines for deep-sea research. D. Technology has made it possible to study the trench without human divers.

5. What is the main purpose of this passage? A. To introduce the Mariana Trench, its formation, conditions, and scientific value. B. To compare the Mariana Trench with Mount Everest. C. To describe how humans first discovered the ocean. D. To encourage tourists to visit the Pacific Ocean.

详细解析

1. 【答案】A

【题型】 细节理解

【定位】 第①段 "It lies in the western Pacific Ocean, east of the Philippines and south of Japan."

【解析】 原文明确指出马里亚纳海沟位于西太平洋，菲律宾以东、日本以南。B 项"大西洋"、C 项"中国沿海"和 D 项"印度洋"均与原文不符。故选 A。

2. 【答案】B

【题型】 细节理解

【定位】 第①段 "Its deepest point, known as the Challenger Deep... reaches about 11 kilometers below sea level."

【解析】 原文说明挑战者深渊最深约达海平面以下 11 公里。A 项"2,550 公里"是海沟长度；C 项"69 公里"是海沟最宽处宽度；D 项"2 公里"指珠峰若放于底部仍距水面 2 公里以上，非深度。故选 B。

3. 【答案】C

【题型】 细节理解

【定位】 第③段 "Yet, even in this harsh environment, some life forms survive... fish, shrimp... that have adapted to the darkness and pressure."

【解析】 原文指出即便环境严酷，仍有生物存活，如鱼、虾等已适应黑暗和高压。A 项"没有生物"与原文相反；B 项"阳光可达底部"与"There is no sunlight"矛盾；D 项"水温很高"与"very cold"矛盾。故选 C。

4. 【答案】D

【题型】 推理判断

【定位】 第④段 "Today, robots and remote-controlled vehicles help scientists study the trench without sending humans down."

【解析】 原文指出如今机器人及遥控潜水器帮助科学家在不派人类下潜的情况下研究海沟，可推断技术使无人类潜水的研究成为可能。A 项"容易且便宜"与"very difficult and expensive"矛盾；B 项"仅一人到达"与 1960 年两人、2012 年 Cameron 矛盾；C 项"不再用潜水艇"文中未明确否定。故选 D。

5. 【答案】A

【题型】 主旨大意

【定位】 全文：第①段介绍位置与深度，第②段形成原因，第③段环境与生物，第④段探索历程，第⑤段科学价值与保护意义。

【解析】 全文从位置、形成、环境、探索与价值等多方面介绍马里亚纳海沟。B 项"与珠峰比较"仅为举例；C 项"人类首次发现海洋"偏离主题；D 项"鼓励旅游"与科普目的不符。故选 A。